APT Overlap (Enterprise): APT28 vs APT29 vs Lazarus Group vs APT41 (Parent + Sub-techniques)

Technique overlap across 4 selected groups | 141 parent T-codes + 173 sub-techniques | campaigns included | software-derived excluded

29 RED
tech + 12 sub
Used by ALL 4 groups

24 ORANGE
tech + 22 sub
Used by exactly 3 of 4 groups

35 YELLOW
tech + 38 sub
Used by exactly 2 of 4 groups

53 LIGHT GREEN
tech + 101 sub
Used by only ONE group

Selected Groups

ID	Group	Aliases	T-codes
G0007	**APT28**	IRON TWILIGHT, SNAKEMACKEREL, Swallowtail, Group 74, Sednit, Sofacy, Pawn Storm, Fancy Bear, STRONTIUM, Tsar Team, Threat Group-4127, TG-4127, Forest Blizzard, FROZENLAKE, GruesomeLarch	77
G0016	**APT29**	IRON RITUAL, IRON HEMLOCK, NobleBaron, Dark Halo, NOBELIUM, UNC2452, YTTRIUM, The Dukes, Cozy Bear, CozyDuke, SolarStorm, Blue Kitsune, UNC3524, Midnight Blizzard	72
G0032	**Lazarus Group**	Labyrinth Chollima, HIDDEN COBRA, Guardians of Peace, ZINC, NICKEL ACADEMY, Diamond Sleet	81
G0096	**APT41**	Wicked Panda, Brass Typhoon, BARIUM	81

Coverage by Tactic

Tactic	RED	ORANGE	YELLOW	LIGHT GREEN	Total
Reconnaissance	0	2	3	2	**7**
Resource Development	2	2	2	1	**7**
Initial Access	2	2	3	2	**9**
Execution	2	3	1	5	**11**
Persistence	4	5	2	4	**15**
Privilege Escalation	3	4	4	1	**12**
Stealth	6	3	5	9	**23**
Defense Impairment	1	2	1	3	**7**
Credential Access	1	2	3	8	**14**
Discovery	2	3	7	9	**21**
Lateral Movement	1	1	0	4	**6**
Collection	3	2	3	3	**11**
Command and Control	6	0	3	4	**13**
Exfiltration	1	1	2	0	**4**
Impact	0	0	1	7	**8**

Technique Matrix (1/2)

indented rows = sub-techniques, each colored by its own overlap

RED (29 + 12 sub) ORANGE (24 + 22 sub) YELLOW (35 + 38 sub) LIGHT GREEN (53 + 101 sub)

Reconnaissance
7 techniques

- **T1589** 3/4 Gather Victim Identity Information
  - .001 Credentials 2/4
  - .002 Email Addres... 1/4
- **T1595** 3/4 Active Scanning
  - .002 Vulnerability ... 3/4
  - .003 Wordlist Scan... 1/4
- **T1591** 2/4 Gather Victim Org Information
  - .004 Identify Roles 1/4
- **T1593** 2/4 Search Open Websites/Domains
  - .001 Social Media 1/4
  - .002 Search Engin... 1/4
- **T1596** 2/4 Search Open Technical Databases
  - .005 Scan Databa... 1/4
- **T1594** 1/4 Search Victim-Owned Websites
- **T1598** 1/4 Phishing for Information
  - .003 Spearphishin... 1/4

Resource Development
7 techniques

- **T1583** 4/4 Acquire Infrastructure
  - .001 Domains 3/4
  - .006 Web Services 3/4
  - .003 Virtual Privat... 1/4
  - .004 Server 1/4
  - .007 Serverless 1/4
- **T1588** 4/4 Obtain Capabilities
  - .002 Tool 4/4
  - .003 Code Signing... 2/4
  - .004 Digital Certific... 1/4
  - .007 Artificial Intelli... 1/4
- **T1584** 3/4 Compromise Infrastructure
  - .001 Domains 2/4
  - .004 Server 1/4
  - .008 Network Devi... 1/4
- **T1586** 3/4 Compromise Accounts
  - .002 Email Accounts 2/4
  - .003 Cloud Accou... 2/4
- **T1585** 2/4 Establish Accounts
  - .001 Social Media ... 2/4
  - .002 Email Accounts 1/4
- **T1587** 2/4 Develop Capabilities
  - .001 Malware 2/4
  - .002 Code Signing... 1/4
  - .003 Digital Certific... 1/4
- **T1608** 1/4 Stage Capabilities
  - .001 Upload Malw... 1/4
  - .002 Upload Tool 1/4

Initial Access
9 techniques

- **T1078** 4/4 Valid Accounts
  - .004 Cloud Accou... 2/4
  - .002 Domain Acco... 1/4
  - .003 Local Accounts 1/4
- **T1566** 4/4 Phishing
  - .001 Spearphishin... 4/4
  - .002 Spearphishin... 2/4
  - .003 Spearphishin... 2/4
- **T1133** 3/4 External Remote Services
- **T1190** 3/4 Exploit Public-Facing Application
- **T1189** 2/4 Drive-by Compromise
- **T1195** 2/4 Supply Chain Compromise
  - .002 Compromise ... 2/4
- **T1199** 2/4 Trusted Relationship
- **T1091** 1/4 Replication Through Removable Media
- **T1669** 1/4 Wi-Fi Networks

Execution
11 techniques

- **T1059** 4/4 Command and Scripting Interpreter
  - .001 PowerShell 4/4
  - .003 Windows Co... 4/4
  - .005 Visual Basic 2/4
  - .004 Unix Shell 1/4
  - .006 Python 1/4
  - .007 JavaScript 1/4
  - .009 Cloud API 1/4
- **T1203** 4/4 Exploitation for Client Execution
- **T1047** 3/4 Windows Management Instrumentation
- **T1053** 3/4 Scheduled Task/Job
  - .005 Scheduled T... 3/4
- **T1204** 3/4 User Execution
  - .001 Malicious Link 3/4
  - .002 Malicious File 3/4
- **T1574** 2/4 Hijack Execution Flow
  - .001 DLL 2/4
  - .006 Dynamic Link... 1/4
  - .013 KernelCallba... 1/4
- **T1106** 1/4 Native API
- **T1197** 1/4 BITS Jobs
- **T1559** 1/4 Inter-Process Communication
  - .002 Dynamic Dat... 1/4
- **T1569** 1/4 System Services
  - .002 Service Exec... 1/4
- **T1651** 1/4 Cloud Administration Command

Persistence
15 techniques

- **T1078** 4/4 Valid Accounts
  - .004 Cloud Accou... 2/4
  - .002 Domain Acco... 1/4
  - .003 Local Accounts 1/4
- **T1098** 4/4 Account Manipulation
  - .002 Additional E... 2/4
  - .001 Additional Cl... 1/4
  - .003 Additional Cl... 1/4
  - .005 Device Regist... 1/4
  - .007 Additional Lo... 1/4
- **T1505** 4/4 Server Software Component
  - .003 Web Shell 3/4
  - .004 IIS Compone... 1/4
- **T1547** 4/4 Boot or Logon Autostart Execution
  - .001 Registry Run ... 4/4
  - .009 Shortcut Mod... 1/4
- **T1037** 3/4 Boot or Logon Initialization Scripts
  - .001 Logon Script ... 1/4
  - .004 RC Scripts 1/4
- **T1053** 3/4 Scheduled Task/Job
  - .005 Scheduled T... 3/4
- **T1133** 3/4 External Remote Services
- **T1542** 3/4 Pre-OS Boot
  - .003 Bootkit 3/4
- **T1546** 3/4 Event Triggered Execution
  - .008 Accessibility ... 2/4
  - .003 Windows Ma... 1/4
  - .015 Component ... 1/4
- **T1136** 2/4 Create Account
  - .001 Local Account 1/4
  - .003 Cloud Account 1/4
- **T1543** 2/4 Create or Modify System Process
  - .003 Windows Ser... 2/4
- **T1112** 1/4 Modify Registry

Privilege Escalation
12 techniques

- **T1078** 4/4 Valid Accounts
  - .004 Cloud Accou... 2/4
  - .002 Domain Acco... 1/4
  - .003 Local Accounts 1/4
- **T1098** 4/4 Account Manipulation
  - .002 Additional E... 2/4
  - .001 Additional Cl... 1/4
  - .003 Additional Cl... 1/4
  - .005 Device Regist... 1/4
  - .007 Additional Lo... 1/4
- **T1547** 4/4 Boot or Logon Autostart Execution
  - .001 Registry Run ... 4/4
  - .009 Shortcut Mod... 1/4
- **T1037** 3/4 Boot or Logon Initialization Scripts
  - .001 Logon Script ... 1/4
  - .004 RC Scripts 1/4
- **T1053** 3/4 Scheduled Task/Job
  - .005 Scheduled T... 3/4
- **T1134** 3/4 Access Token Manipulation
  - .001 Token Impers... 1/4
  - .002 Create Proce... 1/4
- **T1546** 3/4 Event Triggered Execution
  - .008 Accessibility ... 2/4
  - .003 Windows Ma... 1/4
  - .015 Component ... 1/4
- **T1055** 2/4 Process Injection
  - .001 Dynamic-link ... 1/4
- **T1068** 2/4 Exploitation for Privilege Escalation
- **T1484** 2/4 Domain or Tenant Policy Modification
  - .001 Group Policy ... 1/4
  - .002 Trust Modific... 1/4
- **T1543** 2/4 Create or Modify System Process
  - .003 Windows Ser... 2/4
- **T1548** 1/4 Abuse Elevation Control Mechanism
  - .002 Bypass User ... 1/4

Stealth
23 techniques

- **T1027** 4/4 Obfuscated Files or Information
  - .002 Software Pac... 3/4
  - .013 Encrypted/En... 3/4
  - .001 Binary Padding 1/4
  - .003 Steganography 1/4
  - .006 HTML Smugg... 1/4
  - .007 Dynamic API ... 1/4
  - .009 Embedded P... 1/4
- **T1036** 4/4 Masquerading
  - .005 Match Legiti... 4/4
  - .004 Masquerade ... 3/4
  - .003 Rename Legi... 1/4
  - .008 Masquerade ... 1/4
- **T1070** 4/4 Indicator Removal
  - .004 File Deletion 4/4
  - .006 Timestomp 3/4
  - .003 Clear Comma... 2/4
  - .008 Clear Mailbox... 1/4
- **T1078** 4/4 Valid Accounts
  - .004 Cloud Accou... 2/4
  - .002 Domain Acco... 1/4
  - .003 Local Accounts 1/4
- **T1140** 4/4 Deobfuscate/Decode Files or Information
- **T1218** 4/4 System Binary Proxy Execution
  - .011 Rundll32 4/4
  - .005 Mshta 2/4
  - .001 Compiled HT... 1/4
  - .010 Regsvr32 1/4
- **T1134** 3/4 Access Token Manipulation
  - .001 Token Impers... 1/4
  - .002 Create Proce... 1/4
- **T1542** 3/4 Pre-OS Boot
  - .003 Bootkit 3/4
- **T1684** 3/4 Social Engineering
  - .001 Impersonation 3/4
- **T1014** 2/4 Rootkit

Defense Impairment
7 techniques

- **T1685** 4/4 Disable or Modify Tools
  - .005 Clear Windo... 2/4
  - .001 Disable or Mo... 1/4
  - .002 Disable or Mo... 1/4
- **T1553** 3/4 Subvert Trust Controls
  - .002 Code Signing 3/4
  - .005 Mark-of-the-... 1/4
- **T1686** 3/4 Disable or Modify System Firewall
  - .003 Windows Hos... 2/4
- **T1484** 2/4 Domain or Tenant Policy Modification
  - .001 Group Policy ... 1/4
  - .002 Trust Modific... 1/4
- **T1112** 1/4 Modify Registry
- **T1556** 1/4 Modify Authentication Process
  - .007 Hybrid Identity 1/4
- **T1599** 1/4 Network Boundary Bridging

Credential Access
14 techniques

- **T1110** 4/4 Brute Force
  - .003 Password Sp... 3/4
  - .001 Password Gu... 2/4
- **T1003** 3/4 OS Credential Dumping
  - .002 Security Acco... 3/4
  - .001 LSASS Mem... 2/4
  - .003 NTDS 2/4
  - .004 LSA Secrets 1/4
  - .006 DCSync 1/4
- **T1056** 3/4 Input Capture
  - .001 Keylogging 3/4
- **T1528** 2/4 Steal Application Access Token
- **T1555** 2/4 Credentials from Password Stores
  - .003 Credentials fr... 2/4
- **T1557** 2/4 Adversary-in-the-Middle
  - .001 Name Resolu... 1/4
  - .004 Evil Twin 1/4
- **T1040** 1/4 Network Sniffing
- **T1539** 1/4 Steal Web Session Cookie
- **T1552** 1/4 Unsecured Credentials
  - .004 Private Keys 1/4
- **T1556** 1/4 Modify Authentication Process
  - .007 Hybrid Identity 1/4
- **T1558** 1/4 Steal or Forge Kerberos Tickets
  - .003 Kerberoasting 1/4
- **T1606** 1/4 Forge Web Credentials
  - .001 Web Cookies 1/4
  - .002 SAML Tokens 1/4
- **T1621** 1/4 Multi-Factor Authentication Reque...
- **T1649** 1/4 Steal or Forge Authentication Certific...

Discovery
21 techniques

- **T1016** 4/4 System Network Configuration Discovery
  - .001 Internet Conn... 1/4
  - .002 Wi-Fi Discovery 1/4
- **T1083** 4/4 File and Directory Discovery
- **T1057** 3/4 Process Discovery
- **T1087** 3/4 Account Discovery
  - .002 Domain Acco... 3/4
  - .001 Local Account 1/4
  - .004 Cloud Account 1/4
- **T1680** 3/4 Local Storage Discovery
- **T1012** 2/4 Query Registry
- **T1018** 2/4 Remote System Discovery
- **T1033** 2/4 System Owner/User Discovery
- **T1046** 2/4 Network Service Discovery
- **T1049** 2/4 System Network Connections Discovery
- **T1069** 2/4 Permission Groups Discovery
  - .002 Domain Groups 1/4
- **T1082** 2/4 System Information Discovery
- **T1010** 1/4 Application Window Discovery
- **T1040** 1/4 Network Sniffing
- **T1120** 1/4 Peripheral Device Discovery
- **T1124** 1/4 System Time Discovery
- **T1135** 1/4 Network Share Discovery
- **T1482** 1/4 Domain Trust Discovery
- **T1497** 1/4 Virtualization/Sandbox Evasion
  - .001 System Checks 1/4
  - .003 Time Based ... 1/4

Lateral Movement
6 techniques

- **T1021** 4/4 Remote Services
  - .001 Remote Desk... 4/4
  - .002 SMB/Window... 4/4
  - .004 SSH 1/4
  - .006 Windows Re... 1/4
  - .007 Cloud Services 1/4
- **T1550** 3/4 Use Alternate Authentication Material
  - .001 Application A... 2/4
  - .002 Pass the Hash 2/4
  - .003 Pass the Ticket 1/4
  - .004 Web Session... 1/4
- **T1091** 1/4 Replication Through Removable Media
- **T1210** 1/4 Exploitation of Remote Services
- **T1534** 1/4 Internal Spearphishing
- **T1570** 1/4 Lateral Tool Transfer

Collection
11 techniques

- **T1005** 4/4 Data from Local System
- **T1074** 4/4 Data Staged
  - .001 Local Data St... 3/4
  - .002 Remote Data... 2/4
- **T1560** 4/4 Archive Collected Data
  - .001 Archive via Ut... 4/4
  - .003 Archive via C... 2/4
  - .002 Archive via Li... 1/4
- **T1056** 3/4 Input Capture
  - .001 Keylogging 3/4
- **T1213** 3/4 Data from Information Repositories
  - .003 Code Reposit... 2/4
  - .002 Sharepoint 1/4
  - .006 Databases 1/4
- **T1114** 2/4 Email Collection
  - .002 Remote Emai... 2/4
- **T1119** 2/4 Automated Collection
- **T1557** 2/4 Adversary-in-the-Middle
  - .001 Name Resolu... 1/4
  - .004 Evil Twin 1/4
- **T1025** 1/4 Data from Removable Media
- **T1039** 1/4 Data from Network Shared Drive
- **T1113** 1/4 Screen Capture

Command and Control
13 techniques

- **T1001** 4/4 Data Obfuscation
  - .003 Protocol or S... 2/4
  - .001 Junk Data 1/4
  - .002 Steganography 1/4
- **T1071** 4/4 Application Layer Protocol
  - .001 Web Protocols 4/4
  - .002 File Transfer ... 1/4
  - .003 Mail Protocols 1/4
  - .004 DNS 1/4
- **T1090** 4/4 Proxy
  - .001 Internal Proxy 3/4
  - .002 External Proxy 3/4
  - .003 Multi-hop Proxy 2/4
  - .004 Domain Front... 1/4
- **T1102** 4/4 Web Service
  - .002 Bidirectional ... 3/4
  - .001 Dead Drop R... 1/4
- **T1105** 4/4 Ingress Tool Transfer
- **T1573** 4/4 Encrypted Channel
  - .001 Symmetric Cr... 2/4
  - .002 Asymmetric ... 1/4
- **T1008** 2/4 Fallback Channels
- **T1104** 2/4 Multi-Stage Channels
- **T1568** 2/4 Dynamic Resolution
  - .002 Domain Gen... 1/4
- **T1092** 1/4 Communication Through Removable ...
- **T1132** 1/4 Data Encoding
  - .001 Standard Enc... 1/4
- **T1571** 1/4 Non-Standard Port
- **T1665** 1/4 Hide Infrastructure

Exfiltration
4 techniques

- **T1048** 4/4 Exfiltration Over Alternative Protocol
  - .002 Exfiltration O... 2/4
  - .003 Exfiltration O... 2/4
- **T1567** 3/4 Exfiltration Over Web Service
  - .002 Exfiltration to ... 2/4
- **T1030** 2/4 Data Transfer Size Limits
- **T1041** 2/4 Exfiltration Over C2 Channel

Impact
8 techniques

- **T1561** 2/4 Disk Wipe
  - .001 Disk Content ... 2/4
  - .002 Disk Structur... 1/4
- **T1485** 1/4 Data Destruction
- **T1486** 1/4 Data Encrypted for Impact
- **T1489** 1/4 Service Stop
- **T1491** 1/4 Defacement
  - .001 Internal Defa... 1/4
- **T1496** 1/4 Resource Hijacking
  - .001 Compute Hija... 1/4
- **T1498** 1/4 Network Denial of Service
- **T1529** 1/4 System Shutdown/Reboot

Technique Matrix (2/2)

indented rows = sub-techniques, each colored by its own overlap

RED (29 + 12 sub) ORANGE (24 + 22 sub) YELLOW (35 + 38 sub) LIGHT GREEN (53 + 101 sub)

Reconnaissance	Resource Development	Initial Access	Execution	Persistence	Privilege Escalation	Stealth	Defense Impairment	Credential Access	Discovery	Lateral Movement	Collection	Command and Control	Exfiltration	Impact
7 techniques	7 techniques	9 techniques	11 techniques	15 techniques	12 techniques	23 techniques	7 techniques	14 techniques	21 techniques	6 techniques	11 techniques	13 techniques	4 techniques	8 techniques
				T1137 1/4 Office Application Startup		T1055 2/4 Process Injection			T1614 1/4 System Location Discovery					
				.002 Office Test 1/4		.001 Dynamic-link ... 1/4			.001 System Lang... 1/4					
				T1197 1/4 BITS Jobs		T1221 2/4 Template Injection			T1622 1/4 Debugger Evasion					
				T1556 1/4 Modify Authentication Process		T1564 2/4 Hide Artifacts								
				.007 Hybrid Identity 1/4		.001 Hidden Files ... 2/4								
						.003 Hidden Window 1/4								
						T1574 2/4 Hijack Execution Flow								
						.001 DLL 2/4								
						.006 Dynamic Link... 1/4								
						.013 KernelCallba... 1/4								
						T1006 1/4 Direct Volume Access								
						T1197 1/4 BITS Jobs								
						T1202 1/4 Indirect Command Execution								
						T1211 1/4 Exploitation for Stealth								
						T1220 1/4 XSL Script Processing								
						T1480 1/4 Execution Guardrails								
						.001 Environment... 1/4								
						T1497 1/4 Virtualization/Sandbox Evasion								
						.001 System Checks 1/4								
						.003 Time Based ... 1/4								
						T1620 1/4 Reflective Code Loading								
						T1622 1/4 Debugger Evasion								

Technique & Sub-Technique Detail

Overlap	Groups	ID	Technique / Sub-technique	Tactics	APT28	APT29	Lazarus Group	APT41
RED	4/4	T1001	**Data Obfuscation**	Command and Control	✔	✔	✔	✔
YELLOW	2/4	T1001.003	Protocol or Service Impersonation				✔	✔
LIGHT GREEN	1/4	T1001.001	Junk Data		✔			
LIGHT GREEN	1/4	T1001.002	Steganography			✔		
RED	4/4	T1005	**Data from Local System**	Collection	✔	✔	✔	✔
RED	4/4	T1016	**System Network Configuration Discovery**	Discovery	✔	✔	✔	✔
LIGHT GREEN	1/4	T1016.001	Internet Connection Discovery			✔		
LIGHT GREEN	1/4	T1016.002	Wi-Fi Discovery		✔			
RED	4/4	T1021	**Remote Services**	Lateral Movement	✔	✔	✔	✔
RED	4/4	T1021.001	Remote Desktop Protocol		✔	✔	✔	✔
RED	4/4	T1021.002	SMB/Windows Admin Shares		✔	✔	✔	✔
LIGHT GREEN	1/4	T1021.004	SSH				✔	
LIGHT GREEN	1/4	T1021.006	Windows Remote Management			✔		
LIGHT GREEN	1/4	T1021.007	Cloud Services			✔		
RED	4/4	T1027	**Obfuscated Files or Information**	Stealth	✔	✔	✔	✔
ORANGE	3/4	T1027.002	Software Packing			✔	✔	✔
ORANGE	3/4	T1027.013	Encrypted/Encoded File		✔		✔	✔
LIGHT GREEN	1/4	T1027.001	Binary Padding			✔		
LIGHT GREEN	1/4	T1027.003	Steganography			✔		
LIGHT GREEN	1/4	T1027.006	HTML Smuggling			✔		
LIGHT GREEN	1/4	T1027.007	Dynamic API Resolution				✔	
LIGHT GREEN	1/4	T1027.009	Embedded Payloads				✔	
RED	4/4	T1036	**Masquerading**	Stealth	✔	✔	✔	✔
RED	4/4	T1036.005	Match Legitimate Resource Name or Location		✔	✔	✔	✔
ORANGE	3/4	T1036.004	Masquerade Task or Service			✔	✔	✔
LIGHT GREEN	1/4	T1036.003	Rename Legitimate Utilities				✔	
LIGHT GREEN	1/4	T1036.008	Masquerade File Type				✔	
RED	4/4	T1048	**Exfiltration Over Alternative Protocol**	Exfiltration	✔	✔	✔	✔
YELLOW	2/4	T1048.002	Exfiltration Over Asymmetric Encrypted Non-C2 Protocol		✔	✔		
YELLOW	2/4	T1048.003	Exfiltration Over Unencrypted Non-C2 Protocol				✔	✔
RED	4/4	T1059	**Command and Scripting Interpreter**	Execution	✔	✔	✔	✔
RED	4/4	T1059.001	PowerShell		✔	✔	✔	✔
RED	4/4	T1059.003	Windows Command Shell		✔	✔	✔	✔
YELLOW	2/4	T1059.005	Visual Basic			✔	✔	
LIGHT GREEN	1/4	T1059.004	Unix Shell					✔
LIGHT GREEN	1/4	T1059.006	Python			✔		
LIGHT GREEN	1/4	T1059.007	JavaScript					✔
LIGHT GREEN	1/4	T1059.009	Cloud API			✔		
RED	4/4	T1070	**Indicator Removal**	Stealth	✔	✔	✔	✔
RED	4/4	T1070.004	File Deletion		✔	✔	✔	✔
ORANGE	3/4	T1070.006	Timestomp		✔	✔	✔	
YELLOW	2/4	T1070.003	Clear Command History				✔	✔
LIGHT GREEN	1/4	T1070.008	Clear Mailbox Data			✔		
RED	4/4	T1071	**Application Layer Protocol**	Command and Control	✔	✔	✔	✔

Overlap	Groups	ID	Technique / Sub-technique	Tactics	APT28	APT29	Lazarus Group	APT41
RED	4/4	T1071.001	Web Protocols		✔	✔	✔	✔
LIGHT GREEN	1/4	T1071.002	File Transfer Protocols					✔
LIGHT GREEN	1/4	T1071.003	Mail Protocols		✔			
LIGHT GREEN	1/4	T1071.004	DNS					✔
RED	4/4	**T1074**	**Data Staged**	Collection	✔	✔	✔	✔
ORANGE	3/4	T1074.001	Local Data Staging		✔		✔	✔
YELLOW	2/4	T1074.002	Remote Data Staging		✔	✔		
RED	4/4	**T1078**	**Valid Accounts**	Initial Access, Persistence, Privilege Escalation, Stealth	✔	✔	✔	✔
YELLOW	2/4	T1078.004	Cloud Accounts		✔	✔		
LIGHT GREEN	1/4	T1078.002	Domain Accounts			✔		
LIGHT GREEN	1/4	T1078.003	Local Accounts			✔		
RED	4/4	**T1083**	**File and Directory Discovery**	Discovery	✔	✔	✔	✔
RED	4/4	**T1090**	**Proxy**	Command and Control	✔	✔	✔	✔
ORANGE	3/4	T1090.001	Internal Proxy		✔	✔	✔	
ORANGE	3/4	T1090.002	External Proxy		✔	✔	✔	
YELLOW	2/4	T1090.003	Multi-hop Proxy		✔	✔		
LIGHT GREEN	1/4	T1090.004	Domain Fronting			✔		
RED	4/4	**T1098**	**Account Manipulation**	Persistence, Privilege Escalation	✔	✔	✔	✔
YELLOW	2/4	T1098.002	Additional Email Delegate Permissions		✔	✔		
LIGHT GREEN	1/4	T1098.001	Additional Cloud Credentials			✔		
LIGHT GREEN	1/4	T1098.003	Additional Cloud Roles			✔		
LIGHT GREEN	1/4	T1098.005	Device Registration			✔		
LIGHT GREEN	1/4	T1098.007	Additional Local or Domain Groups					✔
RED	4/4	**T1102**	**Web Service**	Command and Control	✔	✔	✔	✔
ORANGE	3/4	T1102.002	Bidirectional Communication		✔	✔	✔	
LIGHT GREEN	1/4	T1102.001	Dead Drop Resolver					✔
RED	4/4	**T1105**	**Ingress Tool Transfer**	Command and Control	✔	✔	✔	✔
RED	4/4	**T1110**	**Brute Force**	Credential Access	✔	✔	✔	✔
ORANGE	3/4	T1110.003	Password Spraying		✔	✔	✔	
YELLOW	2/4	T1110.001	Password Guessing		✔	✔		
RED	4/4	**T1140**	**Deobfuscate/Decode Files or Information**	Stealth	✔	✔	✔	✔
RED	4/4	**T1203**	**Exploitation for Client Execution**	Execution	✔	✔	✔	✔
RED	4/4	**T1218**	**System Binary Proxy Execution**	Stealth	✔	✔	✔	✔
RED	4/4	T1218.011	Rundll32		✔	✔	✔	✔
YELLOW	2/4	T1218.005	Mshta			✔	✔	
LIGHT GREEN	1/4	T1218.001	Compiled HTML File					✔
LIGHT GREEN	1/4	T1218.010	Regsvr32				✔	
RED	4/4	**T1505**	**Server Software Component**	Persistence	✔	✔	✔	✔
ORANGE	3/4	T1505.003	Web Shell		✔	✔		✔
LIGHT GREEN	1/4	T1505.004	IIS Components				✔	
RED	4/4	**T1547**	**Boot or Logon Autostart Execution**	Persistence, Privilege Escalation	✔	✔	✔	✔
RED	4/4	T1547.001	Registry Run Keys / Startup Folder		✔	✔	✔	✔
LIGHT GREEN	1/4	T1547.009	Shortcut Modification				✔	
RED	4/4	**T1560**	**Archive Collected Data**	Collection	✔	✔	✔	✔

Overlap	Groups	ID	Technique / Sub-technique	Tactics	APT28	APT29	Lazarus Group	APT41
RED	4/4	T1560.001	Archive via Utility		✔	✔	✔	✔
YELLOW	2/4	T1560.003	Archive via Custom Method				✔	✔
LIGHT GREEN	1/4	T1560.002	Archive via Library				✔	
RED	4/4	**T1566**	**Phishing**	Initial Access	✔	✔	✔	✔
RED	4/4	T1566.001	Spearphishing Attachment		✔	✔	✔	✔
YELLOW	2/4	T1566.002	Spearphishing Link			✔	✔	
YELLOW	2/4	T1566.003	Spearphishing via Service			✔	✔	
RED	4/4	**T1573**	**Encrypted Channel**	Command and Control	✔	✔	✔	✔
YELLOW	2/4	T1573.001	Symmetric Cryptography		✔		✔	
LIGHT GREEN	1/4	T1573.002	Asymmetric Cryptography					✔
RED	4/4	**T1583**	**Acquire Infrastructure**	Resource Development	✔	✔	✔	✔
ORANGE	3/4	T1583.001	Domains		✔	✔	✔	
ORANGE	3/4	T1583.006	Web Services		✔	✔	✔	
LIGHT GREEN	1/4	T1583.003	Virtual Private Server		✔			
LIGHT GREEN	1/4	T1583.004	Server				✔	
LIGHT GREEN	1/4	T1583.007	Serverless					✔
RED	4/4	**T1588**	**Obtain Capabilities**	Resource Development	✔	✔	✔	✔
RED	4/4	T1588.002	Tool		✔	✔	✔	✔
YELLOW	2/4	T1588.003	Code Signing Certificates				✔	✔
LIGHT GREEN	1/4	T1588.004	Digital Certificates				✔	
LIGHT GREEN	1/4	T1588.007	Artificial Intelligence		✔			
RED	4/4	**T1685**	**Disable or Modify Tools**	Defense Impairment	✔	✔	✔	✔
YELLOW	2/4	T1685.005	Clear Windows Event Logs		✔			✔
LIGHT GREEN	1/4	T1685.001	Disable or Modify Windows Event Log			✔		
LIGHT GREEN	1/4	T1685.002	Disable or Modify Cloud Log			✔		
ORANGE	3/4	**T1003**	**OS Credential Dumping**	Credential Access	✔	✔		✔
ORANGE	3/4	T1003.002	Security Account Manager		✔	✔		✔
YELLOW	2/4	T1003.001	LSASS Memory		✔			✔
YELLOW	2/4	T1003.003	NTDS		✔			✔
LIGHT GREEN	1/4	T1003.004	LSA Secrets			✔		
LIGHT GREEN	1/4	T1003.006	DCSync			✔		
ORANGE	3/4	**T1037**	**Boot or Logon Initialization Scripts**	Persistence, Privilege Escalation	✔	✔		✔
LIGHT GREEN	1/4	T1037.001	Logon Script (Windows)		✔			
LIGHT GREEN	1/4	T1037.004	RC Scripts			✔		
ORANGE	3/4	**T1047**	**Windows Management Instrumentation**	Execution		✔	✔	✔
ORANGE	3/4	**T1053**	**Scheduled Task/Job**	Execution, Persistence, Privilege Escalation		✔	✔	✔
ORANGE	3/4	T1053.005	Scheduled Task			✔	✔	✔
ORANGE	3/4	**T1056**	**Input Capture**	Credential Access, Collection	✔		✔	✔
ORANGE	3/4	T1056.001	Keylogging		✔		✔	✔
ORANGE	3/4	**T1057**	**Process Discovery**	Discovery	✔	✔	✔	
ORANGE	3/4	**T1087**	**Account Discovery**	Discovery		✔	✔	✔
ORANGE	3/4	T1087.002	Domain Account			✔	✔	✔
LIGHT GREEN	1/4	T1087.001	Local Account					✔
LIGHT GREEN	1/4	T1087.004	Cloud Account			✔		

Overlap	Groups	ID	Technique / Sub-technique	Tactics	APT28	APT29	Lazarus Group	APT41
ORANGE	3/4	T1133	External Remote Services	Initial Access, Persistence	✔	✔		✔
ORANGE	3/4	T1134	Access Token Manipulation	Privilege Escalation, Stealth	✔		✔	✔
LIGHT GREEN	1/4	T1134.001	Token Impersonation/Theft		✔			
LIGHT GREEN	1/4	T1134.002	Create Process with Token				✔	
ORANGE	3/4	T1190	Exploit Public-Facing Application	Initial Access	✔	✔		✔
ORANGE	3/4	T1204	User Execution	Execution	✔	✔	✔	
ORANGE	3/4	T1204.001	Malicious Link		✔	✔	✔	
ORANGE	3/4	T1204.002	Malicious File		✔	✔	✔	
ORANGE	3/4	T1213	Data from Information Repositories	Collection	✔	✔		✔
YELLOW	2/4	T1213.003	Code Repositories			✔		✔
LIGHT GREEN	1/4	T1213.002	Sharepoint		✔			
LIGHT GREEN	1/4	T1213.006	Databases					✔
ORANGE	3/4	T1542	Pre-OS Boot	Persistence, Stealth	✔		✔	✔
ORANGE	3/4	T1542.003	Bootkit		✔		✔	✔
ORANGE	3/4	T1546	Event Triggered Execution	Persistence, Privilege Escalation	✔	✔		✔
YELLOW	2/4	T1546.008	Accessibility Features			✔		✔
LIGHT GREEN	1/4	T1546.003	Windows Management Instrumentation Event Subscription			✔		
LIGHT GREEN	1/4	T1546.015	Component Object Model Hijacking		✔			
ORANGE	3/4	T1550	Use Alternate Authentication Material	Lateral Movement	✔	✔		✔
YELLOW	2/4	T1550.001	Application Access Token		✔	✔		
YELLOW	2/4	T1550.002	Pass the Hash		✔			✔
LIGHT GREEN	1/4	T1550.003	Pass the Ticket			✔		
LIGHT GREEN	1/4	T1550.004	Web Session Cookie			✔		
ORANGE	3/4	T1553	Subvert Trust Controls	Defense Impairment		✔	✔	✔
ORANGE	3/4	T1553.002	Code Signing			✔	✔	✔
LIGHT GREEN	1/4	T1553.005	Mark-of-the-Web Bypass			✔		
ORANGE	3/4	T1567	Exfiltration Over Web Service	Exfiltration	✔		✔	✔
YELLOW	2/4	T1567.002	Exfiltration to Cloud Storage				✔	✔
ORANGE	3/4	T1584	Compromise Infrastructure	Resource Development	✔	✔	✔	
YELLOW	2/4	T1584.001	Domains			✔	✔	
LIGHT GREEN	1/4	T1584.004	Server				✔	
LIGHT GREEN	1/4	T1584.008	Network Devices		✔			
ORANGE	3/4	T1586	Compromise Accounts	Resource Development	✔	✔		✔
YELLOW	2/4	T1586.002	Email Accounts		✔	✔		
YELLOW	2/4	T1586.003	Cloud Accounts			✔		✔
ORANGE	3/4	T1589	Gather Victim Identity Information	Reconnaissance	✔	✔	✔	
YELLOW	2/4	T1589.001	Credentials		✔	✔		
LIGHT GREEN	1/4	T1589.002	Email Addresses				✔	
ORANGE	3/4	T1595	Active Scanning	Reconnaissance	✔	✔		✔
ORANGE	3/4	T1595.002	Vulnerability Scanning		✔	✔		✔
LIGHT GREEN	1/4	T1595.003	Wordlist Scanning					✔
ORANGE	3/4	T1680	Local Storage Discovery	Discovery		✔	✔	✔

Overlap	Groups	ID	Technique / Sub-technique	Tactics	APT28	APT29	Lazarus Group	APT41
ORANGE	3/4	T1684	Social Engineering	Stealth	✔		✔	✔
ORANGE	3/4	T1684.001	Impersonation		✔		✔	✔
ORANGE	3/4	T1686	Disable or Modify System Firewall	Defense Impairment	✔	✔	✔	
YELLOW	2/4	T1686.003	Windows Host Firewall		✔		✔	
YELLOW	2/4	T1008	Fallback Channels	Command and Control			✔	✔
YELLOW	2/4	T1012	Query Registry	Discovery			✔	✔
YELLOW	2/4	T1014	Rootkit	Stealth	✔			✔
YELLOW	2/4	T1018	Remote System Discovery	Discovery		✔		✔
YELLOW	2/4	T1030	Data Transfer Size Limits	Exfiltration	✔			✔
YELLOW	2/4	T1033	System Owner/User Discovery	Discovery			✔	✔
YELLOW	2/4	T1041	Exfiltration Over C2 Channel	Exfiltration			✔	✔
YELLOW	2/4	T1046	Network Service Discovery	Discovery			✔	✔
YELLOW	2/4	T1049	System Network Connections Discovery	Discovery			✔	✔
YELLOW	2/4	T1055	Process Injection	Privilege Escalation, Stealth			✔	✔
LIGHT GREEN	1/4	T1055.001	Dynamic-link Library Injection				✔	
YELLOW	2/4	T1068	Exploitation for Privilege Escalation	Privilege Escalation	✔	✔		
YELLOW	2/4	T1069	Permission Groups Discovery	Discovery		✔		✔
LIGHT GREEN	1/4	T1069.002	Domain Groups			✔		
YELLOW	2/4	T1082	System Information Discovery	Discovery			✔	✔
YELLOW	2/4	T1104	Multi-Stage Channels	Command and Control			✔	✔
YELLOW	2/4	T1114	Email Collection	Collection	✔	✔		
YELLOW	2/4	T1114.002	Remote Email Collection		✔	✔		
YELLOW	2/4	T1119	Automated Collection	Collection	✔			✔
YELLOW	2/4	T1136	Create Account	Persistence		✔		✔
LIGHT GREEN	1/4	T1136.001	Local Account					✔
LIGHT GREEN	1/4	T1136.003	Cloud Account			✔		
YELLOW	2/4	T1189	Drive-by Compromise	Initial Access	✔		✔	
YELLOW	2/4	T1195	Supply Chain Compromise	Initial Access		✔		✔
YELLOW	2/4	T1195.002	Compromise Software Supply Chain			✔		✔
YELLOW	2/4	T1199	Trusted Relationship	Initial Access	✔	✔		
YELLOW	2/4	T1221	Template Injection	Stealth	✔		✔	
YELLOW	2/4	T1484	Domain or Tenant Policy Modification	Privilege Escalation, Defense Impairment		✔		✔
LIGHT GREEN	1/4	T1484.001	Group Policy Modification					✔
LIGHT GREEN	1/4	T1484.002	Trust Modification			✔		
YELLOW	2/4	T1528	Steal Application Access Token	Credential Access	✔	✔		
YELLOW	2/4	T1543	Create or Modify System Process	Persistence, Privilege Escalation			✔	✔
YELLOW	2/4	T1543.003	Windows Service				✔	✔
YELLOW	2/4	T1555	Credentials from Password Stores	Credential Access		✔		✔
YELLOW	2/4	T1555.003	Credentials from Web Browsers			✔		✔

Overlap	Groups	ID	Technique / Sub-technique	Tactics	APT28	APT29	Lazarus Group	APT41
YELLOW	2/4	T1557	Adversary-in-the-Middle	Credential Access, Collection	✔		✔	
LIGHT GREEN	1/4	T1557.001	Name Resolution Poisoning and SMB Relay				✔	
LIGHT GREEN	1/4	T1557.004	Evil Twin		✔			
YELLOW	2/4	T1561	Disk Wipe	Impact	✔		✔	
YELLOW	2/4	T1561.001	Disk Content Wipe		✔		✔	
LIGHT GREEN	1/4	T1561.002	Disk Structure Wipe				✔	
YELLOW	2/4	T1564	Hide Artifacts	Stealth	✔		✔	
YELLOW	2/4	T1564.001	Hidden Files and Directories		✔		✔	
LIGHT GREEN	1/4	T1564.003	Hidden Window		✔			
YELLOW	2/4	T1568	Dynamic Resolution	Command and Control		✔		✔
LIGHT GREEN	1/4	T1568.002	Domain Generation Algorithms					✔
YELLOW	2/4	T1574	Hijack Execution Flow	Execution, Stealth			✔	✔
YELLOW	2/4	T1574.001	DLL				✔	✔
LIGHT GREEN	1/4	T1574.006	Dynamic Linker Hijacking					✔
LIGHT GREEN	1/4	T1574.013	KernelCallbackTable				✔	
YELLOW	2/4	T1585	Establish Accounts	Resource Development		✔	✔	
YELLOW	2/4	T1585.001	Social Media Accounts			✔	✔	
LIGHT GREEN	1/4	T1585.002	Email Accounts				✔	
YELLOW	2/4	T1587	Develop Capabilities	Resource Development		✔	✔	
YELLOW	2/4	T1587.001	Malware			✔	✔	
LIGHT GREEN	1/4	T1587.002	Code Signing Certificates				✔	
LIGHT GREEN	1/4	T1587.003	Digital Certificates			✔		
YELLOW	2/4	T1591	Gather Victim Org Information	Reconnaissance	✔		✔	
LIGHT GREEN	1/4	T1591.004	Identify Roles				✔	
YELLOW	2/4	T1593	Search Open Websites/Domains	Reconnaissance			✔	✔
LIGHT GREEN	1/4	T1593.001	Social Media				✔	
LIGHT GREEN	1/4	T1593.002	Search Engines					✔
YELLOW	2/4	T1596	Search Open Technical Databases	Reconnaissance	✔			✔
LIGHT GREEN	1/4	T1596.005	Scan Databases					✔
LIGHT GREEN	1/4	T1006	Direct Volume Access	Stealth	✔			
LIGHT GREEN	1/4	T1010	Application Window Discovery	Discovery			✔	
LIGHT GREEN	1/4	T1025	Data from Removable Media	Collection	✔			
LIGHT GREEN	1/4	T1039	Data from Network Shared Drive	Collection	✔			
LIGHT GREEN	1/4	T1040	Network Sniffing	Credential Access, Discovery	✔			
LIGHT GREEN	1/4	T1091	Replication Through Removable Media	Initial Access, Lateral Movement	✔			
LIGHT GREEN	1/4	T1092	Communication Through Removable Media	Command and Control	✔			
LIGHT GREEN	1/4	T1106	Native API	Execution			✔	
LIGHT GREEN	1/4	T1112	Modify Registry	Persistence, Defense Impairment				✔
LIGHT GREEN	1/4	T1113	Screen Capture	Collection	✔			
LIGHT GREEN	1/4	T1120	Peripheral Device Discovery	Discovery	✔			
LIGHT GREEN	1/4	T1124	System Time Discovery	Discovery			✔	

Overlap	Groups	ID	Technique / Sub-technique	Tactics	APT28	APT29	Lazarus Group	APT41
LIGHT GREEN	1/4	T1132	Data Encoding	Command and Control			✔	
LIGHT GREEN	1/4	T1132.001	Standard Encoding				✔	
LIGHT GREEN	1/4	T1135	Network Share Discovery	Discovery				✔
LIGHT GREEN	1/4	T1137	Office Application Startup	Persistence	✔			
LIGHT GREEN	1/4	T1137.002	Office Test		✔			
LIGHT GREEN	1/4	T1197	BITS Jobs	Execution, Persistence, Stealth				✔
LIGHT GREEN	1/4	T1202	Indirect Command Execution	Stealth			✔	
LIGHT GREEN	1/4	T1210	Exploitation of Remote Services	Lateral Movement	✔			
LIGHT GREEN	1/4	T1211	Exploitation for Stealth	Stealth	✔			
LIGHT GREEN	1/4	T1220	XSL Script Processing	Stealth			✔	
LIGHT GREEN	1/4	T1480	Execution Guardrails	Stealth				✔
LIGHT GREEN	1/4	T1480.001	Environmental Keying					✔
LIGHT GREEN	1/4	T1482	Domain Trust Discovery	Discovery		✔		
LIGHT GREEN	1/4	T1485	Data Destruction	Impact			✔	
LIGHT GREEN	1/4	T1486	Data Encrypted for Impact	Impact				✔
LIGHT GREEN	1/4	T1489	Service Stop	Impact			✔	
LIGHT GREEN	1/4	T1491	Defacement	Impact			✔	
LIGHT GREEN	1/4	T1491.001	Internal Defacement				✔	
LIGHT GREEN	1/4	T1496	Resource Hijacking	Impact				✔
LIGHT GREEN	1/4	T1496.001	Compute Hijacking					✔
LIGHT GREEN	1/4	T1497	Virtualization/Sandbox Evasion	Stealth, Discovery			✔	
LIGHT GREEN	1/4	T1497.001	System Checks				✔	
LIGHT GREEN	1/4	T1497.003	Time Based Checks				✔	
LIGHT GREEN	1/4	T1498	Network Denial of Service	Impact	✔			
LIGHT GREEN	1/4	T1529	System Shutdown/Reboot	Impact			✔	
LIGHT GREEN	1/4	T1534	Internal Spearphishing	Lateral Movement			✔	
LIGHT GREEN	1/4	T1539	Steal Web Session Cookie	Credential Access		✔		
LIGHT GREEN	1/4	T1548	Abuse Elevation Control Mechanism	Privilege Escalation		✔		
LIGHT GREEN	1/4	T1548.002	Bypass User Account Control			✔		
LIGHT GREEN	1/4	T1552	Unsecured Credentials	Credential Access		✔		
LIGHT GREEN	1/4	T1552.004	Private Keys			✔		
LIGHT GREEN	1/4	T1556	Modify Authentication Process	Persistence, Defense Impairment, Credential Access		✔		
LIGHT GREEN	1/4	T1556.007	Hybrid Identity			✔		
LIGHT GREEN	1/4	T1558	Steal or Forge Kerberos Tickets	Credential Access		✔		
LIGHT GREEN	1/4	T1558.003	Kerberoasting			✔		
LIGHT GREEN	1/4	T1559	Inter-Process Communication	Execution	✔			
LIGHT GREEN	1/4	T1559.002	Dynamic Data Exchange		✔			
LIGHT GREEN	1/4	T1569	System Services	Execution				✔
LIGHT GREEN	1/4	T1569.002	Service Execution					✔

Overlap	Groups	ID	Technique / Sub-technique	Tactics	APT28	APT29	Lazarus Group	APT41
LIGHT GREEN	1/4	T1570	Lateral Tool Transfer	Lateral Movement				✔
LIGHT GREEN	1/4	T1571	Non-Standard Port	Command and Control			✔	
LIGHT GREEN	1/4	T1594	Search Victim-Owned Websites	Reconnaissance				✔
LIGHT GREEN	1/4	T1598	Phishing for Information	Reconnaissance	✔			
LIGHT GREEN	1/4	T1598.003	Spearphishing Link		✔			
LIGHT GREEN	1/4	T1599	Network Boundary Bridging	Defense Impairment				✔
LIGHT GREEN	1/4	T1606	Forge Web Credentials	Credential Access		✔		
LIGHT GREEN	1/4	T1606.001	Web Cookies			✔		
LIGHT GREEN	1/4	T1606.002	SAML Tokens			✔		
LIGHT GREEN	1/4	T1608	Stage Capabilities	Resource Development			✔	
LIGHT GREEN	1/4	T1608.001	Upload Malware				✔	
LIGHT GREEN	1/4	T1608.002	Upload Tool				✔	
LIGHT GREEN	1/4	T1614	System Location Discovery	Discovery			✔	
LIGHT GREEN	1/4	T1614.001	System Language Discovery				✔	
LIGHT GREEN	1/4	T1620	Reflective Code Loading	Stealth			✔	
LIGHT GREEN	1/4	T1621	Multi-Factor Authentication Request Generation	Credential Access		✔		
LIGHT GREEN	1/4	T1622	Debugger Evasion	Stealth, Discovery			✔	
LIGHT GREEN	1/4	T1649	Steal or Forge Authentication Certificates	Credential Access		✔		
LIGHT GREEN	1/4	T1651	Cloud Administration Command	Execution		✔		
LIGHT GREEN	1/4	T1665	Hide Infrastructure	Command and Control		✔		
LIGHT GREEN	1/4	T1669	Wi-Fi Networks	Initial Access	✔			